

Federated Learning for Mental Wellbeing Prediction: A Dual Study on Mood Prediction and Cross-Country Generalization

PRANJAL UPADHYAY, IISER Bhopal, India

SAMBIT SAHOO, IISER Bhopal, India

TAASHIF BASHAR, IISER Bhopal, India

In today's world Mental health disorders are a leading cause of disability worldwide. Smartphones are useful sensing devices that are carried by billions of people. With the growth of smartphone-based sensing, large amounts of behavioral, physiological, and contextual data are now available for mental well-being prediction. However, centralizing such sensitive data raises privacy and security concerns. Federated Learning (FL) offers a decentralized alternative that trains models locally on user devices. This report presents a comprehensive two-part study on wellbeing prediction using FL. **Part I** focuses on mood *classification* using multimodal sensor data from the DiversityOne dataset, comparing deep architectures such as LSTM, StaticNet (MLP), and TabM in both centralized and federated settings. **Part II** extends the analysis to a *regression* paradigm, studying cross-country generalization (Jilin → London/Copenhagen) using FL and centralized baselines. The combined results highlight the challenges of non-IID data, client drift, domain shift, and cross-cultural heterogeneity. We conclude with FL training methods suitable for real-world wellbeing applications.

ACM Reference Format:

Pranjal Upadhyay, Sambit Sahoo, and Taashif Bashar. 2025. Federated Learning for Mental Wellbeing Prediction: A Dual Study on Mood Prediction and Cross-Country Generalization. *ACM Trans. Graph.* 37, 4, Article 111 (August 2025), 7 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 INTRODUCTION

Problem Addressed. Modern mental wellbeing prediction relies heavily on smartphone and wearable sensor data, which captures fine-grained behavioral patterns such as mobility, interaction, social context, and device usage. Training accurate machine learning models on such data typically requires large centralized datasets. However, centralizing sensitive wellbeing data introduces significant security, ethical, and privacy concerns. This work addresses the core problem of developing accurate machine learning and deep learning models for wellbeing prediction *without* centralized data collection by leveraging Federated Learning (FL).

Motivation and Importance. Mental health disorders represent a global public health crisis, and early detection is essential for intervention. FL enables privacy-preserving computation while allowing

models to benefit from population-scale behavioral data. Additionally, wellbeing varies significantly across cultural, demographic, and geographic contexts, making cross-country generalization an important challenge. Understanding when and why FL succeeds or fails is crucial for real-world deployment of wellbeing analytics applications.

Gap in Existing Literature. Prior works have mostly tried mental health prediction using centralized methods and they often rely on small datasets, assume mild non-IID distributions, and ignore the severe domain shifts that occur between populations. Many works also focus on classification tasks alone, lacking multimodal analysis across both classification and regression paradigms. Critically, few studies compare centralized and federated learning end-to-end on large, real-world multi-country datasets. Existing works rarely address realistic deployment challenges such as incremental data accumulation on user devices and cascading FL, where models must learn under cold-start conditions as data gradually becomes available.

Our Contributions. This report addresses the gaps through a two-part technical analysis:

- (1) A detailed classification study using LSTM, StaticNet, and TabM models by taking each user as FL clients from the DiversityOne dataset.
- (2) An empirical investigation of model failure modes in FL: client drift, mode collapse, and non-IID divergence.
- (3) robust regression study testing generalization capabilities across three distinct populations: Jilin (China), London (UK), and Copenhagen (Denmark).
- (4) Development and evaluation of “Cascading FL” to simulate cold-start scenarios.
- (5) Development of a web interface for visualizing FL vs. Centralized performance, running live training simulations, and performing real-time inference of mood on user-provided data.

2 RELATED WORK

We summarize relevant literature across FL, mobile sensing, and mental health analytics:

Summary of Literature (Per-Paper: What's Done and What's Missing)

- **FedTherapist (2023):** *Done:* Uses FL on linguistic expressions (smartphone typing + voice) for privacy-preserving mental health monitoring. *Missing:* Limited dataset size and no evaluation under strong non-IID or cross-population variation, does not explore incremental data accumulation.

Authors' addresses: Pranjal Upadhyay, pranjal22@iiserb.ac.in, IISER Bhopal, Bhopal, India; Sambit Sahoo, iiserb.bhopal, India, sambit22@iiserb.ac.in; Taashif Bashar, iiserb.bhopal, Bhopal, India, taashif22@iiserb.ac.in.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM 0730-0301/2025/8-ART111

<https://doi.org/XXXXXXX.XXXXXXX>

- **FedMood (2021):** *Done:* Proposes a multi-view FL framework using keystroke dynamics and accelerometer data for depression detection. *Missing:* Small user base (20 users), assumes mild non-IID settings, and does not analyze FL failure modes like client drift, cross country transferability, cascading FL.
- **Healthcare Analytics (2024):** Healthcare Analytics *Done:* Provides a comprehensive survey of mental-health prediction models using centralized machine learning on clinical data, social media posts, audio, surveys, and limited wearable/device logs. Uses traditional ML (SVM, RF, LR, CNN, LSTM) on small datasets (20–100 participants), primarily within single countries and without real-world deployment considerations. *Missing:* No FL setup, no large-scale multimodal smartphone sensing, no benchmarking of modern deep models, no cross-country generalization, and no cascading / cold-start simulations.
- **Dubey et al. (2025):** *Done:* Proposes a multimodal CNN–LSTM FL framework using physiological signals and online activity, with differential privacy and homomorphic encryption for secure training. *Missing:* Uses a small synthetic multimodal dataset, no large-scale client simulation, no analysis of FL failure modes, no non-IID stress tests, no cross-country generalization, and no cascading / cold-start FL.
- **Healthcare FL Survey (Kim et al., 2024):** *Done:* Provides an overview of FL in medical imaging, EHR analytics, and distributed diagnostic pipelines. *Missing:* Lacks empirical work on multimodal smartphone sensing, non-IID behavioral data, or cross-cultural mental wellbeing prediction.

PART I: CLASSIFICATION STUDY (DIVERSITYONE DATASET)

3 EXPERIMENTAL DESIGN & METHODOLOGY

This experiment evaluates the efficacy of various neural network architectures for predicting user mood based on passive smartphone sensing data. Unlike previous multi-country studies, this experiment isolates the **Trento (Italy)** subset of the DiversityOne dataset to establish a rigorous baseline within a homogeneous cultural context. We implemented a unified training pipeline to compare Centralized Learning (CL) against Federated Learning (FL), specifically addressing the challenges of Class Imbalance and Client Drift.

3.1 The Trento Dataset

To minimize cross-cultural noise variables, we utilized data exclusively from the Trento deployment site.

- **Participants:** 213 Unique Users.
- **Data Density:** Each user contributed approximately 700 to 800 aggregated data points (30-minute time bins).
- **Target Variable:** Self-reported Mood (Valence), mapped to a discrete 5-class scale (0: Very Negative to 4: Very Positive).

3.2 Feature Engineering

We aggregated sensor streams into **30-minute bins**, resulting in 26 passive sensing features per timestamp:

- **App Usage:** Unique apps, event counts, usage entropy, and notification frequency.
- **Device Dynamics:** Battery levels, charging status, ringer modes, and screen interactions.
- **Physical Activity:** Accelerometer-derived activity counts (walking, still, in-vehicle) and step contexts.

3.3 Data Partitioning Strategy

To ensure a fair evaluation of generalization capabilities, the data was partitioned chronologically for every user:

- **Training Set (70%):** Used for model optimization.
- **Validation Set (10%):** Used for hyperparameter tuning and early stopping logic.
- **Test Set (20%):** Strictly held out for final inference.

3.4 Unified Pipeline Architecture

A core contribution of this study is the development of a modularcor, "Write Once, Train Anywhere" pipeline. This architecture ensures that comparisons between Centralized and Federated performance are topologically identical.

- **Model Injection:** Neural network architectures are defined in a standalone module ('model.py') and imported into both the Centralized trainer and the Federated Client. This guarantees that parameter counts and forward-pass logic are identical across paradigms.
- **State Persistence:** Whether trained centrally or via Flower simulation, final model weights are serialized into a standard '.pth' state dictionary to eliminate "implementation gap" errors.
- **Inference:** After training, saved model weights can be used to classify the test or new data. This is used to deploy the model.

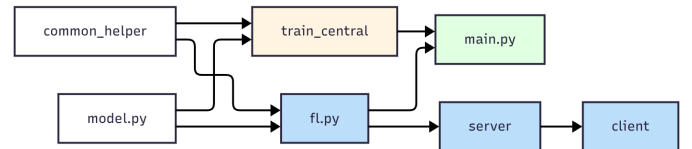


Fig. 1. Pipeline Architecture: Model: Code of models, Main: Use trained weights to classify new data.

3.5 Architecture Definitions

We implemented three distinct architectures targeting different data modalities (Sequential vs. Static).

1. TimeSeriesLSTM (Sequential). Designed to capture temporal dependencies, this model processes a sequence of $T = 10$ timesteps (approx. 5 hours). It utilizes a standard LSTM layer followed by a linear head on the last hidden state.

$$h_t = \text{LSTM}(x_t, h_{t-1}); \quad \hat{y} = W_{\text{head}}(h_T) + b$$

- **Input:** (Batch, 10, 26)
- **Hidden Size:** 64

2. StaticNet (Deep MLP). A baseline deep Multi-Layer Perceptron processing instantaneous features. It uses Batch Normalization (BN) and Dropout to prevent overfitting on the static feature vector.

Input₂₆ $\xrightarrow{128}$ BN/ReLU/Drop(0.3) $\xrightarrow{64}$ BN/ReLU/Drop(0.3) $\xrightarrow{\text{Output}}$

3. TabM (Tabular Ensemble). A state-of-the-art architecture designed for tabular data. It utilizes *LinearReLUEmbeddings* to project continuous features into a higher-dimensional space ($d = 8$) and processes them through an internal ensemble ($k = 32$).

$$\hat{y} = \frac{1}{k} \sum_{i=1}^k \text{Backbone}_i(\text{Embed}(x))$$

The model computes the mean output across the ensemble dimension during the forward pass, providing superior robustness without increasing inference latency significantly.

3.6 Addressing Class Imbalance

The dataset exhibits severe class imbalance, with users predominantly reporting "Neutral" or "Slightly Positive" moods. Naive training resulted in **Mode Collapse**, where models predicted the majority class with near-zero F1-scores for minority classes.

To mitigate this, we implemented **Inverse Frequency Weighting**. The loss calculated for rare classes (e.g., "Very Negative") was multiplied by a high penalty factor (up to 22.0x), while the majority class loss was down-weighted (0.35x). This forced the models to learn features distinguishing the minority classes.

4 RESULTS (PART I)

4.1 Centralized Baseline Results

We trained three distinct architectures (LSTM, StaticNet, TabM) in a centralized manner (pooling all 213 users' data) for 100 Epochs using the weighted loss function.

Table 1. Centralized classification performance on the Trento Test Set.

Model Architecture	Accuracy	Verdict
TabM (Tabular Ensemble)	69.95%	Best Performer
LSTM (Sequential)	68.28%	Competitive
StaticNet (Deep MLP)	56.11%	Baseline

TabM Superiority: The Tabular Ensemble (TabM) outperformed sequential models, suggesting that learning rich embeddings for sensor values is more valuable than modeling temporal history for this specific task.

4.2 Analysis: Time Series vs. Static Data

A key component of this experiment was explicitly analyzing the trade-off between leveraging temporal history (Time Series) versus employing deeper feature extraction on current data (Static).

Temporal Context (LSTM). Our LSTM model utilized a sliding window of $T = 10$ data points, representing approximately **5 hours of past user history**. This design was intended to capture sequential dependencies and mood transitions over time.

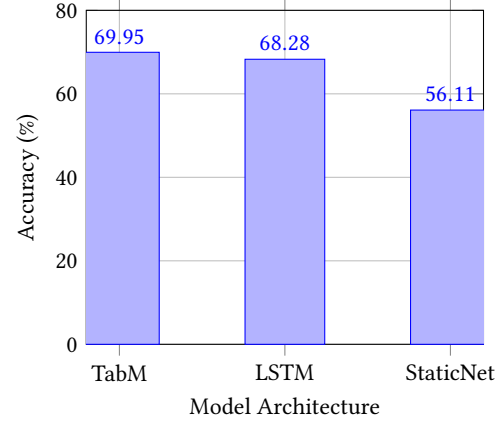


Fig. 2. Centralized Test Accuracy by Architecture.

Static Feature Extraction (TabM). In contrast, the TabM model operates on "static" data—it utilizes only the current sensor snapshot without any access to previous states. Instead of temporal history, it relies on rigorous feature extraction and learnable embeddings to maximize information yield.

Conclusion. Our results indicate that while Time Series modeling (LSTM) is significantly better than simple static baselines (MLP), a State-of-the-Art (SOTA) feature extractor (TabM) can surpass time-series approaches.

- **Feature Depth vs. History:** The superior performance of TabM suggests that the immediate sensor snapshot contains sufficient signal to predict mood if the feature extraction is deep enough; historical data is helpful, but not strictly necessary.
- **Deployment Trade-off:** Although the feature-extractor model (TabM) is computationally heavier per inference step than the LSTM, it offers a distinct operational advantage: it does not require a "warm-up" buffer of 5 hours of past data. Once trained, it can classify mood instantaneously based on current sensor readings.

4.3 Federated Learning Results

We deployed the models into a Federated Simulation using 213 clients (one per user) utilizing Federated Averaging (FedAvg).

Stability Threshold (Local Epochs). A critical experiment determined the optimal amount of local training computation (E).

- **1 Local Epoch:** Stable but slow convergence.
- **3 Local Epochs (Optimal):** Yielded the best performance, allowing models to learn local features without over-optimizing to user bias.
- **>3 Local Epochs:** Performance degraded rapidly due to **Client Drift**. Models diverged into disjoint local minima, and averaging these weights destroyed the global structure.

Final Federated Performance. The Federated performance was evaluated using Cross-Entropy Loss (lower is better), as class imbalance makes accuracy a less granular metric for stability analysis.

Table 2. Federated Learning Performance (Cross-Entropy Loss).

Model	Loss (Lower is Better)	Analysis
TabM	1.5	Most Robust
LSTM	1.9	Moderate Stability
StaticNet	2.2	Poor Convergence

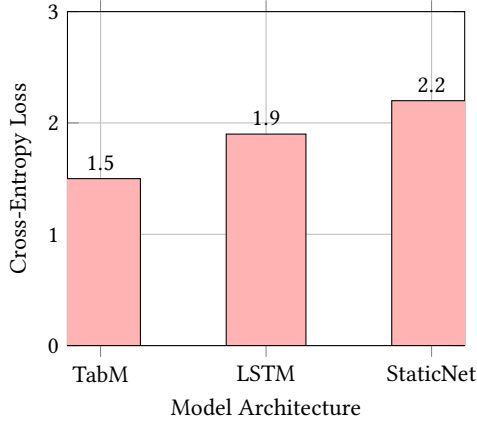


Fig. 3. Federated Learning Loss (Lower indicates better stability).

Result: TabM proved most robust in the Federated setting (Loss 1.5). Its embedding layers allowed it to generalize across heterogeneous clients better than the LSTM (Loss 1.9), which struggled to aggregate diverse temporal timelines, and StaticNet (Loss 2.2), which failed to find a common global minimum.

PART II: REGRESSION AND CROSS-COUNTRY GENERALIZATION

5 METHODOLOGY (PART II)

5.1 Dataset

We use three populations:

- Jilin
- London
- Copenhagen

Features include:

- Static features (Big Five personality, demographics)
- Contextual features (activity, location, social context)
- Sensor features (steps, screen-on duration)

5.2 Model Architecture

We implemented a Multi-Layer Perceptron (MLP) regressor to capture non-linear sensor-mood relationships. The architecture processes ≈ 45 features (including encoded context) through three dense layers with ReLU activations:

$$\text{Input} \xrightarrow{\text{ReLU}} 128 \xrightarrow{\text{ReLU}} 64 \xrightarrow{\text{ReLU}} 32 \rightarrow \text{Output}$$

5.3 Experimental Setup

Simulations were conducted in PyTorch with users modeled as distinct clients. We compared Federated Learning (SGD optimizer) against Centralized baselines (Adam optimizer).

Parameter	Value
Rounds / Local Epochs	50-100 / 5
Batch Size	32
Optimizers	SGD ($\eta = 0.01$, FL), Adam ($\eta = 0.001$, Cent.)
Loss / Aggregation	MSE / FedAvg

Table 3. Training Hyperparameters

Protocol: Gradient clipping was applied to mitigate local client drift. "Cascading" experiments simulated cold-starts by incrementally releasing data (Day 1 $\rightarrow$ N) to mimic real-world accumulation.

6 RESULTS (PART II)

6.1 Intra-Country Performance

We evaluated the convergence behavior of Federated Learning compared to Centralized baselines across all three datasets. While FL models successfully converged, they exhibited higher initial volatility due to the non-IID nature of local client data.

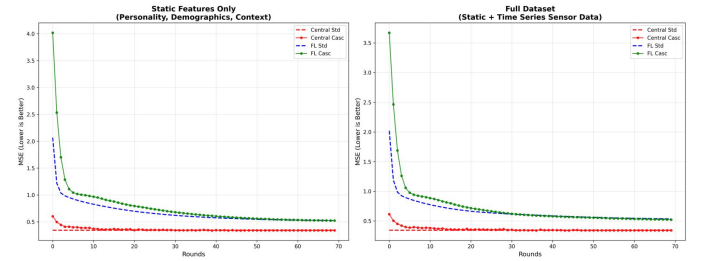


Fig. 4. Jilin: Comparison of Centralized vs. FL Convergence.

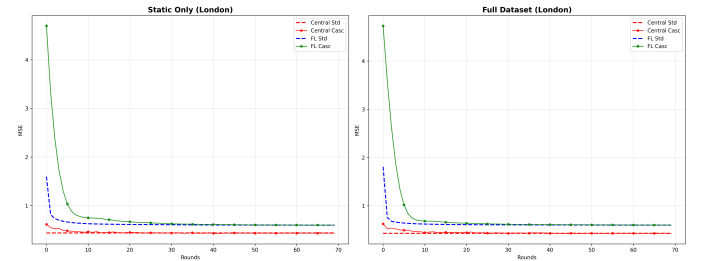


Fig. 5. London: Comparison of Centralized vs. FL Convergence.

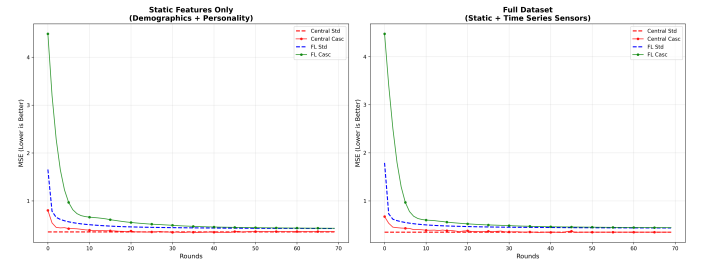


Fig. 6. Copenhagen: Comparison of Centralized vs. FL Convergence.

6.2 Cross-Country Generalization (Jilin → London/Copenhagen)

Testing models trained on Jilin against European datasets revealed significant challenges regarding domain shift:

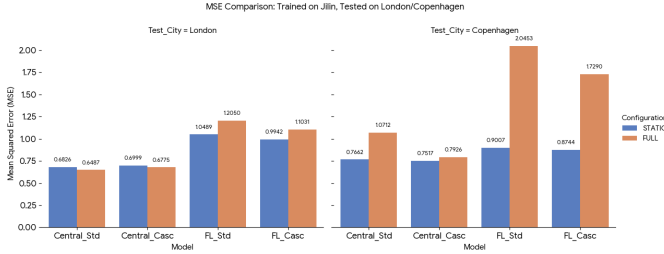


Fig. 7. Cross-Country Generalization: MSE Comparison of Jilin-trained models tested on London and Copenhagen.

- **London:** The Centralized model generalized moderately well, while FL retained reasonable error margins.
- **Copenhagen:** The FL Cascading model failed dramatically ($MSE > 1.7$), indicating a severe distribution shift that standard aggregation (FedAvg) could not bridge.

6.3 Analysis

- **Domain Shift Sensitivity:** FL suffers more from domain shift than centralized training because it aggregates model weights rather than raw data, losing the nuance needed to generalize across diverse cultures.
- **Cultural Divergence:** The failure on the Copenhagen dataset suggests distinct behavioral patterns in Denmark compared to China (Jilin), which the model viewed as "out-of-distribution" noise.
- **Cascading Limitations:** While the "Cascading" setup effectively simulated cold-start dynamics, it could not solve the fundamental distribution mismatch between populations.

7 LIMITATIONS

- Minority mood classes were difficult to learn, even after weighting, due to limited per-user samples.
- Computationally heavy models like TabM are challenging to federate efficiently on resource-limited devices.
- The extreme non-IID nature of user behavioral data caused instability in FL training and led to mode collapse under FedAvg.
- Cascading FL helped simulate real-world deployment but still struggled with data scarcity during early rounds.

8 CONCLUSION

In this project, we presented a comprehensive two-part study combining large-scale classification and cross-country regression for wellbeing prediction using the DiversityOne dataset. Our work demonstrated the value of multimodal smartphone sensing and systematically evaluated three deep architectures—LSTM, StaticNet, and TabM—under both centralized and federated settings. We

also built a fully functional web application that enables interactive model visualization, real-time FL training, and intuitive comparison between centralized, standard FL, and Cascading FL. This tool not only strengthens reproducibility but also showcases how federated systems could be deployed in real-world scenarios.

Across our experiments, the centralized TabM model emerged as the strongest classifier, showing the importance of rich feature embeddings for static behavioral data. The regression study further highlighted how behavioral, contextual, and personality features can be combined to predict mood at a continuous level. The Cascading FL setup, implemented in both our codebase and the web interface, successfully simulated realistic cold-start conditions and provided important insights into how FL systems behave when users contribute data incrementally. Together, these contributions offer a robust foundation for privacy-preserving wellbeing analytics.

9 FUTURE WORK

Future works aim to incorporate the weekly survey component of the DiversityOne dataset to enabling the model to learn longer-term behavioral trends and try to put attention mechanisms to identify which factors persist across weeks in longer time period. Such longitudinal modeling may reveal deeper emotional patterns beyond short-term fluctuations. Additionally, we aim to integrate a personalized feedback module within the web application so that users can receive adaptive, data-driven recommendations to support their mental wellbeing.

Pranjal	Sambit	Taashif
• Preprocessed device usage data to formulate logic for count and average metrics, which were subsequently utilized by Sambit for feature extraction. • Initially made models with synchronous interactions data, which are weekly self-report questionnaires, before transitioning to diachronic data to incorporate time-series analysis. • Conducted a comprehensive literature review on classification models for tabular data, exploring classic machine learning, deep learning for time series, SOTA transformers, and feature embedding techniques. • Designed a data pipeline to transform feature-extracted data into user-level partitions, enabling standardized data sharing for both federated and centralized setups. • Architected the complete Federated Learning simulation and performed the architectural design for specific neural network models, including TabM, LSTM, and MLP. • Developed a unified training pipeline that allows centralized and federated setups to share a common model definition, ensuring that saved models can be directly utilized by both setups for inference. • Conducted rigorous experiments on model architectures to determine the most suitable approach, comparing time-series versus static approaches, feature extraction, class imbalance, and model drift. • Tested the extremes of the pipeline to check when it will fail, resulting in a detailed comparative benchmark of centralized versus federated learning performance.	Conducted literature review of the papers, finding out the recent works and missing parts, performed exploratory data analysis on the DiversityOne dataset, and carried out feature selection and extraction for preparing the federated dataset for the classification problem (using app usage, device usage, and motion streams) using the timeditaries surveys(30-minute window) . Implemented entropy-based features. Also attempted initial FL experiments on the dataset using smaller models, though the results were not satisfactory due to instability.	Implemented the full regression pipeline (from data extraction to training) and conducted comprehensive experiments: • Pipeline Design: Implemented the complete regression workflow for mood prediction, moving from raw sensor logs to aggregated features. Implemented a Multi-Layer Perceptron (MLP) regressor architecture to capture non-linear relationships in behavioral data. • FL vs. Centralized Benchmarking: Conducted comparative studies between Centralized Learning and Federated Learning to quantify the privacy-utility trade-off. • Cascading FL Simulation: Implemented a “Cascading” experimental protocol to simulate real-world deployment conditions. This involved training models on incrementally available data to evaluate model performance . • Classification Task Contribution: Assisted in the classification study. • Cross-Country Generalization: Led the evaluation of domain shift by testing model transferability across diverse populations. Specifically, trained models on the Jilin cohort and performed zero-shot inference on London and Copenhagen datasets. • Interactive Web Application: Developed a web interface that serves two key functions: – Visualization: Renders real-time comparative plots of training convergence (MSE loss). – Inference: Enables users to input their own behavioral metrics to receive instantaneous mental health score predictions.

Table 4. Work division among team members.

10 REFERENCES

REFERENCES

- [1] Ramanathan, V., et al. *DiversityOne: A Large-Scale Multimodal Smartphone Sensing Dataset for Mental Wellbeing*. arXiv preprint arXiv:2502.03347, 2025.
- [2] Wang, R., et al. *StudentLife: Assessing Mental Health, Academic Performance and Behavioral Trends of College Students*. Available at: <https://studentlife.cs.dartmouth.edu/dataset.html>, 2014.
- [3] Bonawitz, K., et al. Practical Secure Aggregation for Federated Learning on User-Held Data. *ACM CCS*, 2017.
- [4] Abadi, M., et al. Deep Learning with Differential Privacy. *ACM CCS*, 2016.
- [5] Kim, J., et al. Survey of Medical Applications of Federated Learning. *Healthcare Informatics Research*, 2024.
- [6] Li, X., et al. Slim-Sense: Efficient WiFi Sensing for ISAC Systems. *IEEE Transactions on Mobile Computing*, 2024.
- [7] Zhang, Y., et al. FedTherapist: Mental Health Monitoring with Smartphones via Federated Learning. arXiv:2310.16538, 2023.
- [8] Lin, Y., et al. FedMood: Federated Learning Framework for Personalized Depression Detection. 2023.
- [9] Dubey, S., et al. A Multimodal CNN-LSTM Federated Learning Framework for Secure Mental Health Prediction with Differential Privacy. 2025.